1. Description

1.1. Project

Project Name	cubeMX
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	05/20/2026

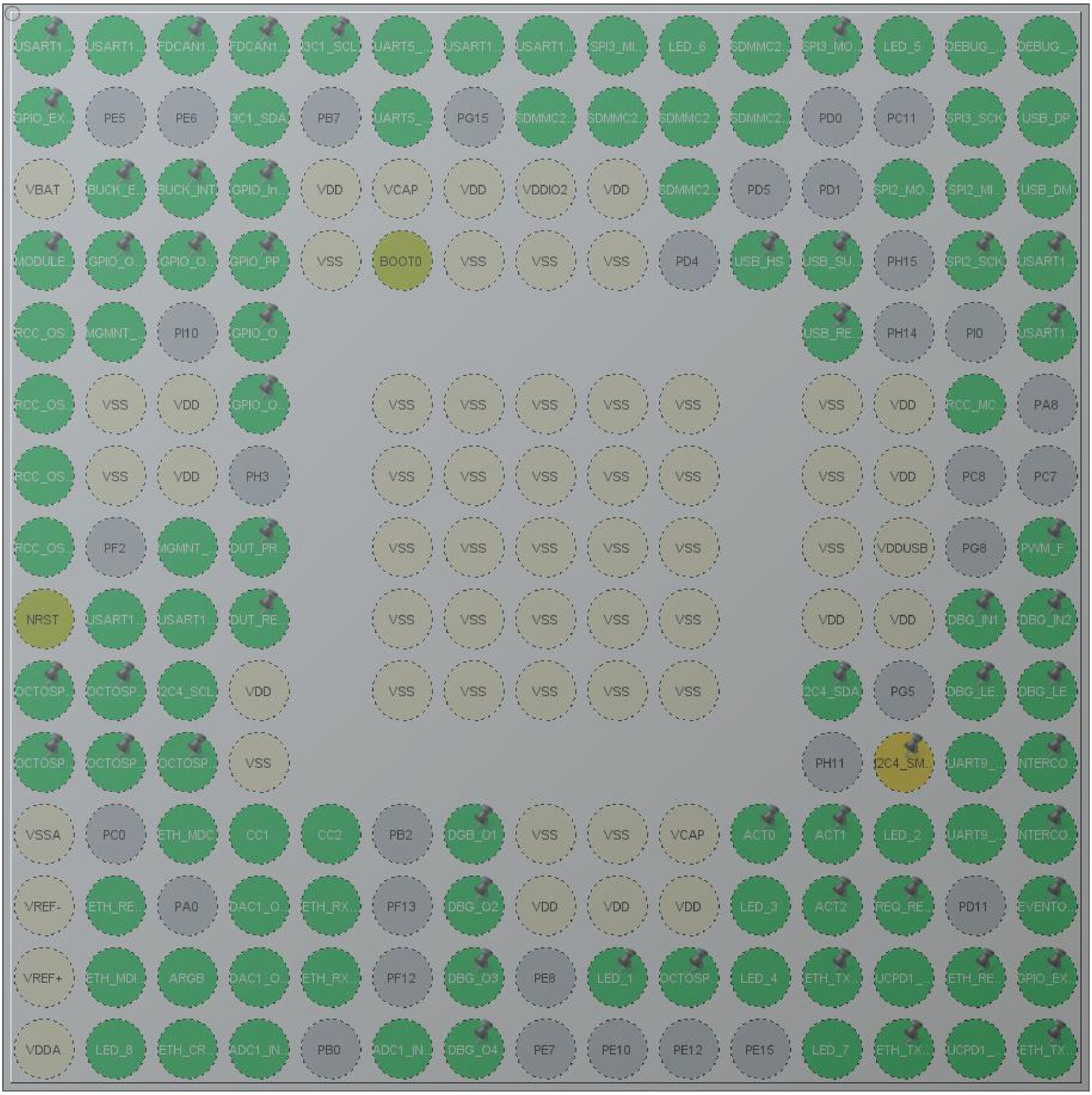
1.2. MCU

MCU Series	STM32H5
MCU Line	STM32H563/H573
MCU name	STM32H563IIKx
MCU Package	UFBGA176
MCU Pin number	201

1.3. Core(s) information

Core(s)	ARM Cortex-M33

2. Pinout Configuration



UFBGA176 +25 (Top view)

3. Pins Configuration

Pin Number UFBGA176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	PE3	I/O	USART10_TX	
A2	PE2	I/O	USART10_RX	
A3	PE1	I/O	FDCAN1_TX	
A4	PE0	I/O	FDCAN1_RX	
A5	PB8	I/O	I3C1_SCL	
A6	PB5	I/O	UART5_RX	
A7	PG14	I/O	USART10_RTS	
A8	PG13	I/O	USART10_CTS	
A9	PB4(NJTRST)	I/O	SPI3_MISO	
A10	PB3(JTDO/TRACESWO)	I/O	TIM2_CH2	LED_6
A11	PD7	I/O	SDMMC2_CMD	
A12	PC12	I/O	SPI3_MOSI	
A13	PA15(JTDI)	I/O	TIM2_CH1	LED_5
A14	PA14(JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
A15	PA13(JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
B1	PE4	I/O	GPIO_EXTI4	
B4	PB9	I/O	I3C1_SDA	
B6	PB6	I/O	UART5_TX	
B8	PG12	I/O	SDMMC2_D3	
B9	PG11	I/O	SDMMC2_D2	
B10	PG10	I/O	SDMMC2_D1	
B11	PD6	I/O	SDMMC2_CK	
B14	PC10	I/O	SPI3_SCK	
B15	PA12	I/O	USB_DP	
C1	VBAT	Power		
C2	PI7 *	I/O	GPIO_Output	BUCK_ENABLE
C3	PI6	I/O	GPIO_EXTI6	BUCK_INT
C4	PI5 *	I/O	GPIO_Input	
C5	VDD	Power		
C6	VCAP	Power		
C7	VDD	Power		
C8	VDDIO2	Power		
C9	VDD	Power		
C10	PG9	I/O	SDMMC2_D0	
C13	PI3	I/O	SPI2_MOSI	
C14	PI2	I/O	SPI2_MISO	

Pin Number UFBGA176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
C15	PA11	I/O	USB_DM	
D1	PC13 *	I/O	GPIO_Input	MODULE_PRESENT
D2	PI8 *	I/O	GPIO_Output	GPIO_OD_1
D3	PI9 *	I/O	GPIO_Output	
D4	PI4 *	I/O	GPIO_Input	GPIO_PP_2
D5	VSS	Power		
D6	BOOT0	Boot		
D7	VSS	Power		
D8	VSS	Power		
D9	VSS	Power		
D11	PD3 *	I/O	GPIO_Input	USB_HS
D12	PD2 *	I/O	GPIO_Input	USB_SUSP
D14	PI1	I/O	SPI2_SCK	
D15	PA10	I/O	USART1_RX	
E1	PC14-OSC32_IN(OSC32_IN)	I/O	RCC_OSC32_IN	
E2	PF0	I/O	I2C2_SDA	MGMNT_SDA
E4	PI11 *	I/O	GPIO_Input	GPIO_OD_IN1
E12	PH13 *	I/O	GPIO_Output	USB_RESET
E15	PA9	I/O	USART1_TX	
F1	PC15-OSC32_OUT(OSC32_OUT)	I/O	RCC_OSC32_OUT	
F2	VSS	Power		
F3	VDD	Power		
F4	PH2 *	I/O	GPIO_Input	GPIO_OD_IN2
F6	VSS	Power		
F7	VSS	Power		
F8	VSS	Power		
F9	VSS	Power		
F10	VSS	Power		
F12	VSS	Power		
F13	VDD	Power		
F14	PC9	I/O	RCC_MCO_2	
G1	PH0-OSC_IN(PH0)	I/O	RCC_OSC_IN	
G2	VSS	Power		
G3	VDD	Power		
G6	VSS	Power		
G7	VSS	Power		
G8	VSS	Power		
G9	VSS	Power		

Pin Number UFBGA176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
G10	VSS	Power		
G12	VSS	Power		
G13	VDD	Power		
H1	PH1-OSC_OUT(PH1)	I/O	RCC_OSC_OUT	
H3	PF1	I/O	I2C2_SCL	MGMNT_SCL
H4	PH4 *	I/O	GPIO_Input	DUT_PRESENT
H6	VSS	Power		
H7	VSS	Power		
H8	VSS	Power		
H9	VSS	Power		
H10	VSS	Power		
H12	VSS	Power		
H13	VDDUSB	Power		
H15	PC6	I/O	TIM3_CH1	PWM_FAN
J1	NRST	Reset		
J2	PF3	I/O	USART11_TX	
J3	PF4	I/O	USART11_RX	
J4	PH5 *	I/O	GPIO_Output	DUT_RESET
J6	VSS	Power		
J7	VSS	Power		
J8	VSS	Power		
J9	VSS	Power		
J10	VSS	Power		
J12	VDD	Power		
J13	VDD	Power		
J14	PG7 *	I/O	GPIO_Input	DBG_IN1
J15	PG6 *	I/O	GPIO_Input	DBG_IN2
K1	PF7	I/O	OCTOSPI1_IO2	
K2	PF6	I/O	OCTOSPI1_IO3	
K3	PF5	I/O	I2C4_SCL	
K4	VDD	Power		
K6	VSS	Power		
K7	VSS	Power		
K8	VSS	Power		
K9	VSS	Power		
K10	VSS	Power		
K12	PH12	I/O	I2C4_SDA	
K14	PG4 *	I/O	GPIO_Output	DBG_LED1
K15	PG3 *	I/O	GPIO_Output	DBG_LED2

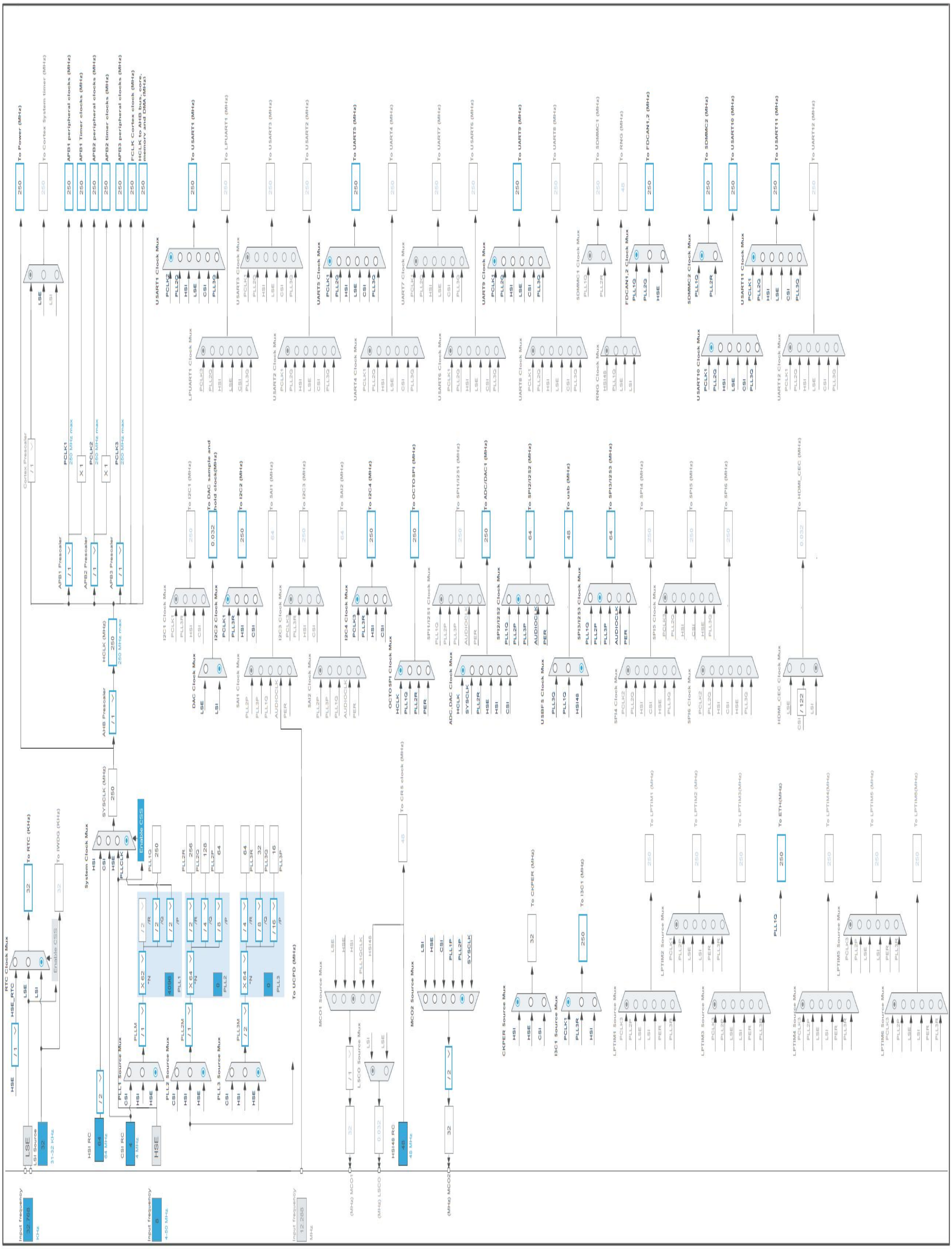
Pin Number UFBGA176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
L1	PF10	I/O	OCTOSPI1_CLK	
L2	PF9	I/O	OCTOSPI1_IO1	
L3	PF8	I/O	OCTOSPI1_IO0	
L4	VSS	Power		
L13	PH10 **	I/O	I2C4_SMBA	
L14	PD15	I/O	UART9_TX	
L15	PG2 *	I/O	GPIO_Output	INTERCOM_OUT
M1	VSSA	Power		
M3	PC1	I/O	ETH_MDC	
M4	PC2	I/O	ADC1_INP12	CC1
M5	PC3	I/O	ADC1_INP13	CC2
M7	PG1 *	I/O	GPIO_Output	DGB_O1
M8	VSS	Power		
M9	VSS	Power		
M10	VCAP	Power		
M11	PH6 *	I/O	GPIO_Input	ACT0
M12	PH8 *	I/O	GPIO_Input	ACT1
M13	PH9	I/O	TIM1_CH2	LED_2
M14	PD14	I/O	UART9_RX	
M15	PD13 *	I/O	GPIO_Input	INTERCOM_IN
N1	VREF-	Power		
N2	PA1	I/O	ETH_REF_CLK	
N4	PA4	I/O	DAC1_OUT1	
N5	PC4	I/O	ETH_RXD0	
N7	PG0 *	I/O	GPIO_Output	DBG_O2
N8	VDD	Power		
N9	VDD	Power		
N10	VDD	Power		
N11	PE13	I/O	TIM1_CH3	LED_3
N12	PH7 *	I/O	GPIO_Input	ACT2
N13	PD12 *	I/O	GPIO_Input	REQ_RESET
N15	PD10 *	I/O	EVENTOUT	
P1	VREF+	Power		
P2	PA2	I/O	ETH_MDIO	
P3	PA6	I/O	TIM13_CH1	ARGB
P4	PA5	I/O	DAC1_OUT2	
P5	PC5	I/O	ETH_RXD1	
P7	PF15 *	I/O	GPIO_Output	DBG_O3
P9	PE9	I/O	TIM1_CH1	LED_1

Pin Number UFBGA176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
P10	PE11	I/O	OCTOSPI1_NCS	
P11	PE14	I/O	TIM1_CH4	LED_4
P12	PB12	I/O	ETH_TXD0	
P13	PB13	I/O	UCPD1_CC1	
P14	PD9 *	I/O	GPIO_Output	ETH_RESET
P15	PD8	I/O	GPIO_EXTI8	
R1	VDDA	Power		
R2	PA3	I/O	TIM2_CH4	LED_8
R3	PA7	I/O	ETH_CRS_DV	
R4	PB1	I/O	ADC1_INP5	
R6	PF11	I/O	ADC1_INP2	
R7	PF14 *	I/O	GPIO_Output	DBG_O4
R12	PB10	I/O	TIM2_CH3	LED_7
R13	PB11	I/O	ETH_TX_EN	
R14	PB14	I/O	UCPD1_CC2	
R15	PB15	I/O	ETH_TXD1	

* The pin is affected with an I/O function

** The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H5
Line	STM32H563/H573
MCU	STM32H563IIKx
Datasheet	DS00000_Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

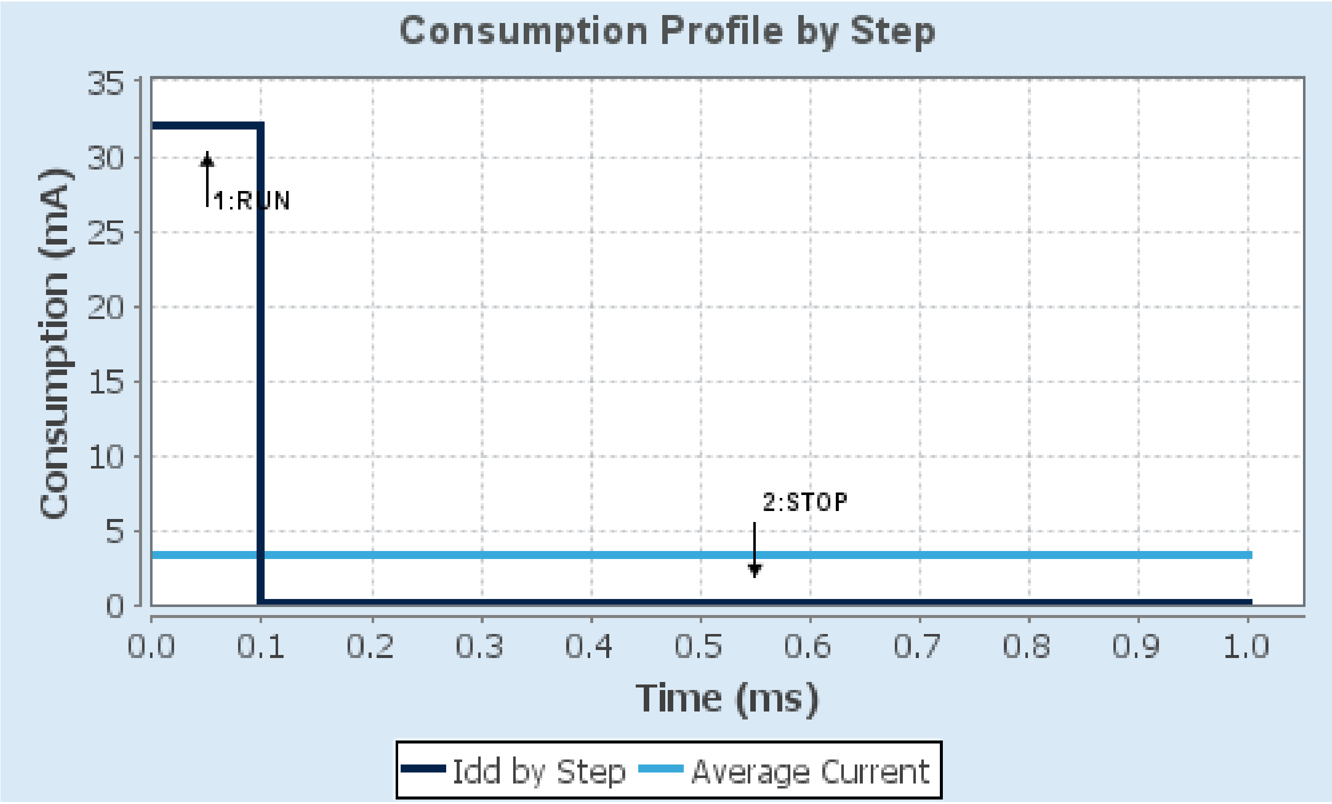
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0	SVOS5: System-Scale5/SMPS
Fetch Type	FLASH_ON/Cache2Ways_ALL_RAM_RETENTION	Flash-PwrDwn_PwrDwnStop_OFF
CPU Frequency	250 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL_CLOCKS_OFF
Clock Source Frequency	8 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	32 mA	51.5 µA
Duration	0.1 ms	0.9 ms
DMIPS	535.0	0.0
Ta Max	121.35	124.99
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.25 mA
Battery Life	1 month, 13 days, 4 hours	Average DMIPS	535.0 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
35
30
25
20
15
10
5
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	cubeMX
Project Folder	C:\dev\EtherBenchProbe\cubeMX
Toolchain / IDE	EWARM V9.20
Firmware Package Name and Version	STM32Cube FW_H5 V1.5.1
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_ADC1_Init	ADC1
4	MX_CORDIC_Init	CORDIC
5	MX_CRC_Init	CRC
6	MX_DAC1_Init	DAC1
7	MX_DCACHE1_Init	DCACHE1
8	MX_ETH_Init	ETH
9	MX_FDCAN1_Init	FDCAN1
10	MX_FMAC_Init	FMAC
11	MX_I2C2_Init	I2C2

Rank	Function Name	Peripheral Instance Name
12	MX_I2C4_Init	I2C4
13	MX_I3C1_Init	I3C1
14	MX_ICACHE_Init	ICACHE
15	MX_OCTOSPI1_Init	OCTOSPI1
16	MX_RTC_Init	RTC
17	MX_SDMMC2_SD_Init	SDMMC2
18	MX_SPI2_Init	SPI2
19	MX_SPI3_Init	SPI3
20	MX_TIM1_Init	TIM1
21	MX_TIM2_Init	TIM2
22	MX_TIM3_Init	TIM3
23	MX_TIM13_Init	TIM13
24	MX_UART5_Init	UART5
25	MX_UART9_Init	UART9
26	MX_USART1_UART_Init	USART1
27	MX_USART10_UART_Init	USART10
28	MX_USART11_UART_Init	USART11
29	MX_UCPD1_Init	UCPD1
30	MX_USB_PCD_Init	USB

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN2: IN2 Single-ended

IN5: IN5 Single-ended

IN12: IN12 Single-ended

mode: IN13

mode: Temperature Sensor Channel

mode: Vrefint Channel

3.1.1. Parameter Settings:

ADCs_Common_Settings:	
Mode	Independent mode
ADC_Settings:	
Clock Prescaler	Asynchronous clock mode divided by 4
Resolution	ADC 12-bit resolution
Scan Conversion Mode	Disabled
Data Alignment	Right alignment
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled
ADC_Regular_ConversionMode:	
Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Sampling Mode	Normal
Rank	1
Channel	Channel 2
Sampling Time	2.5 Cycles
Offset Number	No offset
Monitored by	None
ADC_Injected_ConversionMode:	
Enable Injected Conversions	Disable
Analog Watchdog 1:	

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.2. BOOTPATH
mode: Activated

3.3. CORDIC
mode: Activated

3.4. CRC
mode: Activated
3.4.1. Parameter Settings:

Basic Parameters:

Default Polynomial State	Enable
Default Init Value State	Enable

Advanced Parameters:

Input Data Inversion Mode	None
Output Data Inversion Mode	Disable
Input Data Format	Bytes

3.5. DAC1
OUT1 connected to: only external pin
OUT2 connected to: only external pin
3.5.1. Parameter Settings:

Common DAC Settings:

DAC High Frequency Mode	Disable

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
DMA double data mode	Disable
Signed Format	Disable

Trigger	None
User Trimming	Factory trimming

DAC Out2 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
DMA double data mode	Disable
Signed Format	Disable
Trigger	None
User Trimming	Factory trimming

3.6. DCACHE1
mode: Activated
3.6.1. Parameter Settings:

Basic Parameters:

DCACHE Read Burst Type	DCACHE READ BURST WRAP

3.7. DEBUG
Debug: Serial Wire

3.8. ETH
Mode: RMII
3.8.1. Parameter Settings:

General : Ethernet Configuration:

Ethernet MAC Address	00:80:E1:00:00:00
Tx Descriptor Length	4
Rx Descriptor Length	4
Rx Buffers Length	1524

3.9. FDCAN1
mode: Activated
3.9.1. Parameter Settings:

Basic Parameters:

Clock Divider	Divide kernel clock by 1
Frame Format	Classic mode
Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	1
Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1
Std Filters Nbr	0
Ext Filters Nbr	0
Tx Fifo Queue Mode	FIFO mode
Bit Timings Parameters:	
Nominal Prescaler	16
Nominal Time Quantum	**64.0 ***
Nominal Time Seg1	1
Nominal Time Seg2	1
Nominal Time for one Bit	**192 ***
Nominal Baud Rate	**5208333 ***

3.10. FMAC
mode: Mode

3.11. I2C2
I2C: I2C
3.11.1. Parameter Settings:

Timing configuration:	
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x60808CD3 ***
Slave Features:	
Clock No Stretch Mode	Disabled

General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.12. I2C4

I2C: I2C

3.12.1. Parameter Settings:

Timing configuration:

I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x60808CD3 ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.13. I3C1

Mode: Controller

3.13.1. Parameter Settings:

Hardware Specific Timing Constraints:

SDA Rise Time (ns)	350

Bus Characteristics:

Bus usage	**Mixed I3C and I2C ***
Frequency I3C Controller	1250
Frequency I2C Controller	1000
I3C Duty Cycle	**4 ***
I2C Duty Cycle	30
SCL OD Freq	**1351 ***

SCL PP Freq	**1250 ***
Timing Register 0	**0x4aae09bd ***

Bus Timing Activity:

SDA Hold Time	SDA Hold 0.5 multiple of Kernel Clock Period
Wait Time	Own Controller Activity state 0
Bus Free Duration (ns)	**1050 ***
Bus Idle Duration (ns)	200000
Timing Register 1	**0x008200f8 ***

Basic Configuration:

Target Hot Join Acknowledgement	Disable
Stall Configuration	
- Stall Time (ns)	0
- ACK Stalling	Disable
- CCC T-Bit Stalling	Disable
- Tx Parity Stalling	Disable
- Rx Parity Stalling	Disable
Timing Register 2	0x00000000

Advanced Configuration:

Own Dynamic Address	0
SDA High Keeper	Disable
Fifo Configuration	
- Transmit Fifo Threshold	Threshold 1 byte
- Receive Fifo Threshold	Threshold 1 byte
- Control Fifo	Disable
- Status Fifo	Disable

3.14. ICACHE
mode: Memory address remap
Mode: 2-ways set associative cache
3.14.1. Parameter Settings:

Region 0:

Region	Disable

Region 1:

Region	Disable

Region 2:

Region	Disable

Region 3:

Region	Disable

3.15. MEMORYMAP
mode: Activated

3.16. OCTOSPI1
Mode: Quad SPI
Data: IO[3:0]
3.16.1. Parameter Settings:

Generic:

Fifo Threshold	1
Memory Mode	Single
Memory Type	Micron
Memory Size	16 bits (2 Byte)
Chip Select High Time	1
Free Running Clock	Disable
Clock Mode	Low
Wrap Size	Not Supported
Clock Prescaler	0
Sample Shifting	None
Delay Hold Quarter Cycle	Disable
Chip Select Boundary	Disable
Refresh Rate	0

3.17. PWR
mode: Power saving mode
mode: Privilege attributes
3.17.1. PWR Privilege :

Privilege PWR:

PWR Privilege	Disable

3.18. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator
Low Speed Clock (LSE) : Crystal/Ceramic Resonator
mode: Master Clock Output 2

3.18.1. RCC Privilege :

Privilege RCC:

Privilege of RCC Non-Secure Items	Disable

3.18.2. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	5 WS (6 CPU cycle)
Flash Programming Delay	2

RCC Parameters:

HSI Calibration Value	64
CSI Calibration Value	32
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
TIM Prescaler Selection	Disabled

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

PLL1/2/3 Parameters:

PLL1 input frequency range	Between 8 and 16 MHz
PLL2 input frequency range	Between 8 and 16 MHz

3.19. RTC

mode: Activate Clock Source
mode: Activate Calendar

3.19.1. Parameter Settings:

General:

Hour Format	Hourformat 24
Asynchronous Predivider value	127
Synchronous Predivider value	255
Bin Mode	Free running BCD calender mode

Calendar Time:

Data Format	BCD data format

Hours	0
Minutes	0
Seconds	0
Day Light Saving: value of hour adjustment	Daylightsaving None
Store Operation	Storeoperation Reset

Calendar Date:

Week Day	Monday
Month	January
Date	1
Year	0

3.19.2. RTC Privilege:

Privilege RTC:

RTC full privilege	Disable

Backup register:

Start zone 1	RTC_BKP_DR0
Start Zone 2	RTC_BKP_DR0
start zone 3	RTC_BKP_DR0

Privilege Backup register :

Backup Register PrivZone	Non-privilege

Privilege RTC Feature:

RTC Initialisation	Non-Privilege
RTC Alarm A	Non-Privilege
RTC Alarm B	Non-Privilege
RTC Calibration	Non-Privilege
RTC TimeStamp	Non-Privilege
RTC WakeUpTimer	Non-Privilege

3.20. SDMMC2
Mode: SD 4 bits Wide bus
3.20.1. Parameter Settings:

SDMMC parameters:

Clock transition on which the bit capture is made	Rising transition
SDMMC Clock output enable when the bus is idle	Disable the power save for the clock
SDMMC hardware flow control	The hardware control flow is disabled
SDMMC clock divide factor	0
Is external transceiver present ?	no

3.21. SPI2

Mode: Full-Duplex Slave

3.21.1. Parameter Settings:

Basic Parameters:	
Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First
Clock Parameters:	
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
CRC Parameters:	
CRC Calculation	Disabled
Advanced Parameters:	
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled
Ready Signal Polarity	High

3.22. SPI3

Mode: Full-Duplex Master

3.22.1. Parameter Settings:

Basic Parameters:	
Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	2
Baud Rate	**32.0 MBits/s ***
Clock Polarity (CPOL)	Low

Clock Phase (CPHA)	1 Edge

CRC Parameters:

CRC Calculation	Disabled

Advanced Parameters:

NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled
Ready Master Management	Internal
Ready Signal Polarity	High

3.23. SYS
Timebase Source: TIM6

3.24. TIM1
Clock Source : Internal Clock
Channel1: PWM Generation CH1
Channel2: PWM Generation CH2
Channel3: Output Compare CH3
Channel4: PWM Generation CH4

3.24.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

Output Compare Channel 3:

Mode	Frozen (used for Timing base)
Pulse (16 bits value)	0
Output compare preload	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 4:

Mode	PWM mode 1

Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.25. TIM2
Clock Source : Internal Clock
Channel1: PWM Generation CH1
Channel2: PWM Generation CH2
Channel3: PWM Generation CH3
Channel4: PWM Generation CH4
3.25.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (32 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (32 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (32 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (32 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.26. TIM3

Clock Source : Internal Clock

Channel1: PWM Generation CH1

3.26.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.27. TIM13

mode: Activated

Channel1: PWM Generation CH1

3.27.1. Parameter Settings:

Counter Settings:	
Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable
Clear Input:	
Clear Input Source	Disable
PWM Generation Channel 1:	
Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.28. UART5

Mode: Asynchronous

3.28.1. Parameter Settings:

Basic Parameters:	
Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1
Advanced Parameters:	
Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration
Advanced Features:	
Auto Baudrate	Disable

TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.29. UART9
Mode: Asynchronous
3.29.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.30. UCPD1
UCPD Mode: Sink

3.30.1. Parameter Settings:

Version	1.0

3.31. USART1
Mode: Asynchronous
3.31.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.32. USART10
Mode: Asynchronous
Hardware Flow Control (RS232): CTS/RTS
3.32.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.33. USART11
Mode: Asynchronous
3.33.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.34. USB

Mode: Device_Only

3.34.1. Parameter Settings:

Basic Parameters:

Speed	Full Speed 12MBit/s
Physical interface	Internal Phy
Signal start of frame	Disabled

Power Parameters:

Low Power	Disabled
Link Power Management	Disabled
Battery Charging	Disabled

EndPoint Parameters:

Bulk double buffer	Disabled
Iso single buffer	Disabled

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC2	ADC1_INP12	Analog mode	No pull-up and no pull-down	n/a	CC1
	PC3	ADC1_INP13	Analog mode	No pull-up and no pull-down	n/a	CC2
	PB1	ADC1_INP5	Analog mode	No pull-up and no pull-down	n/a	
	PF11	ADC1_INP2	Analog mode	No pull-up and no pull-down	n/a	
DAC1	PA4	DAC1_OUT1	Analog mode	No pull-up and no pull-down	n/a	
	PA5	DAC1_OUT2	Analog mode	No pull-up and no pull-down	n/a	
DEBUG	PA14(JTCK/ SWCLK)	DEBUG_JTCK- SWCLK	n/a	n/a	n/a	
	PA13(JTMS/ SWDIO)	DEBUG_JTMS- SWDIO	n/a	n/a	n/a	
ETH	PC1	ETH_MDC	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PA1	ETH_REF_CLK	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PC4	ETH_RXD0	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PA2	ETH_MDIO	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PC5	ETH_RXD1	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PB12	ETH_TXD0	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PA7	ETH_CRS_DV	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PB11	ETH_TX_EN	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PB15	ETH_TXD1	Alternate Function Push Pull	No pull-up and no pull-down	High	
FDCAN1	PE1	FDCAN1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE0	FDCAN1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
I2C2	PF0	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	MGMNT_SDA
	PF1	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	MGMNT_SCL
I2C4	PF5	I2C4_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PH12	I2C4_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
I3C1	PB8	I3C1_SCL	Alternate Function Push Pull	Pull-up	Very High	
	PB9	I3C1_SDA	Alternate Function Push Pull	Pull-up	Very High	
OCTOSPI1	PF7	OCTOSPI1_IO2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF6	OCTOSPI1_IO3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF10	OCTOSPI1_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF9	OCTOSPI1_IO1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF8	OCTOSPI1_IO0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE11	OCTOSPI1_NCS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
RCC	PC14-OSC32_IN(OSC32_IN)	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT(OSC32_O	RCC_OSC32_OUT	n/a	n/a	n/a	
	PC9	RCC_MCO_2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PH0-OSC_IN(PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT(PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
SDMMC2	PD7	SDMMC2_CMD	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PG12	SDMMC2_D3	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PG11	SDMMC2_D2	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PG10	SDMMC2_D1	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PD6	SDMMC2_CK	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PG9	SDMMC2_D0	Alternate Function Push Pull	No pull-up and no pull-down	High	
SPI2	PI3	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PI2	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PI1	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
SPI3	PB4(NJTRST)	SPI3_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC12	SPI3_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC10	SPI3_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM1	PH9	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	LED_2
	PE13	TIM1_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	LED_3
	PE9	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	LED_1
	PE14	TIM1_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	LED_4
TIM2	PB3(JTDO/TRACESWO)	TIM2_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	LED_6
	PA15(JTDI)	TIM2_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	LED_5
	PA3	TIM2_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	LED_8
	PB10	TIM2_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	LED_7
TIM3	PC6	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	PWM_FAN
TIM13	PA6	TIM13_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	ARGB
UART5	PB5	UART5_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB6	UART5_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
UART9	PD15	UART9_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD14	UART9_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
UCPD1	PB13	UCPD1_CC1	Analog mode	No pull-up and no pull-down	n/a	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PB14	UCPD1_CC2	Analog mode	No pull-up and no pull-down	n/a	
USART1	PA10	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA9	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART10	PE3	USART10_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE2	USART10_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG14	USART10_RTS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG13	USART10_CTS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART11	PF3	USART11_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PF4	USART11_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USB	PA12	USB_DP	n/a	n/a	n/a	
	PA11	USB_DM	n/a	n/a	n/a	
Single Mapped Signals	PH10	I2C4_SMBA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
GPIO	PE4	GPIO_EXTI4	**External Interrupt Mode with Falling edge trigger detection**	No pull-up and no pull-down	n/a	
	PI7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	BUCK_ENABLE
	PI6	GPIO_EXTI6	**External Interrupt Mode with Falling edge trigger detection**	No pull-up and no pull-down	n/a	BUCK_INT
	PI5	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	
	PC13	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	MODULE_PRESENT
	PI8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO_OD_1
	PI9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PI4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO_PP_2
	PD3	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USB_HS
	PD2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USB_SUSP
	PI11	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO_OD_IN1
	PH13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	USB_RESET
	PH2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	GPIO_OD_IN2
	PH4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	DUT_PRESENT
	PH5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DUT_RESET
	PG7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	DBG_IN1
	PG6	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	DBG_IN2
	PG4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DBG_LED1
	PG3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DBG_LED2
	PG2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	INTERCOM_OUT
	PG1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DGB_O1
	PH6	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	ACT0

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PH8	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	ACT1
	PD13	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	INTERCOM_IN
	PG0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DBG_O2
	PH7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	ACT2
	PD12	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	REQ_RESET
	PD10	EVENTOUT	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PF15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DBG_O3
	PD9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ETH_RESET
	PD8	GPIO_EXTI8	**External Interrupt Mode with Falling edge trigger detection**	No pull-up and no pull-down	n/a	
	PF14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DBG_O4

4.2. GPDMA1

4.3. GPDMA2

4.4. LINKEDLIST

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
TIM6 global interrupt	true	0	0
Flash non-secure global interrupt	unused		
RCC non-secure global interrupt	unused		
EXTI Line4 interrupt	unused		
EXTI Line6 interrupt	unused		
EXTI Line8 interrupt	unused		
ADC1 global interrupt	unused		
DAC1 interrupt	unused		
FDCAN1 interrupt 0	unused		
FDCAN1 interrupt 1	unused		
TIM1 Break interrupt	unused		
TIM1 Update interrupt	unused		
TIM1 Trigger and Commutation interrupts	unused		
TIM1 Capture Compare interrupt	unused		
TIM2 global interrupt	unused		
TIM3 global interrupt	unused		
I2C2 Event interrupt	unused		
I2C2 Error interrupt	unused		
SPI2 global interrupt	unused		
SPI3 global interrupt	unused		
USART1 global interrupt	unused		
UART5 global interrupt	unused		
USB FS global interrupt	unused		
UCPD1 global interrupt	unused		
OCTOSPI1 global interrupt	unused		
USART10 global interrupt	unused		
USART11 global interrupt	unused		
UART9 global interrupt	unused		
SDMMC2 global interrupt	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
FPU global interrupt	unused		
Instruction cache global interrupt	unused		
Data cache global interrupt	unused		
Ethernet global interrupt	unused		
Ethernet Wakeup global interrupt	unused		
CORDIC global interrupt	unused		
FMAC global interrupt	unused		
TIM13 global interrupt	unused		
I3C1 Event interrupt	unused		
I3C1 Error interrupt	unused		
I2C4 Event interrupt	unused		
I2C4 Error interrupt	unused		

4.5.2. NVIC Code generation

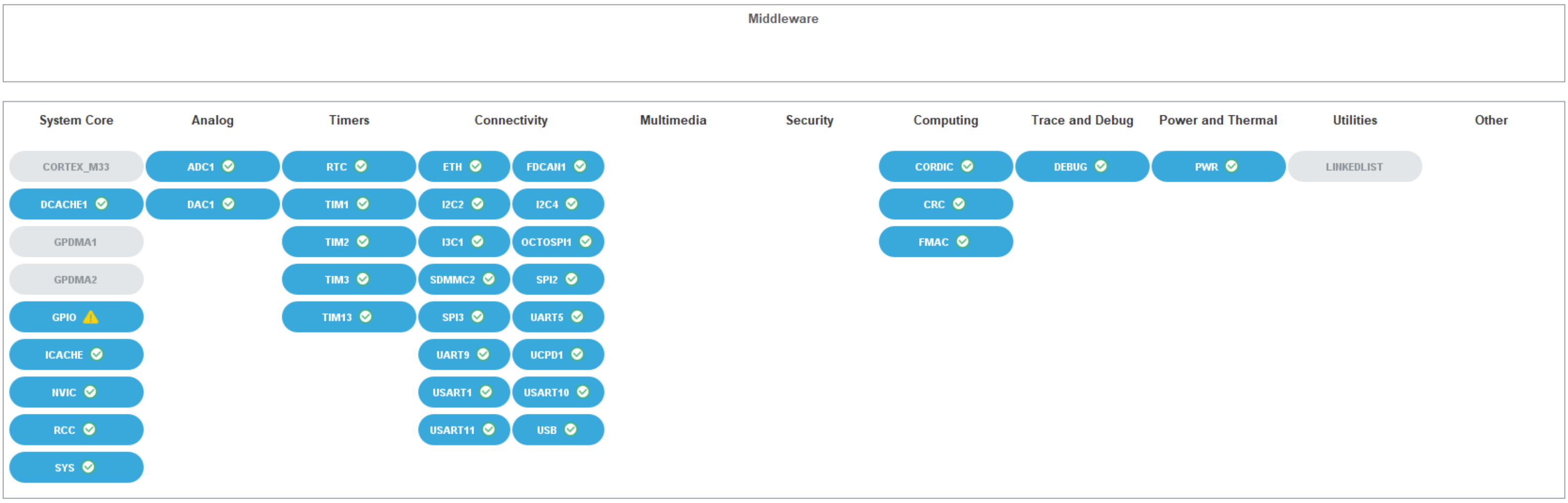
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
TIM6 global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current



6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h5-bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h5-ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h5-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h5-series-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/faq-security-stm32h5.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/secure-manager-introduction-v1.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h5.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1474-security-bulletin-tn1474stpsirt-information-on-softwarebased--microarchitectural-timing-sidechannel-attacks-on-mcus-with-trustzone-for--armv8m-

	stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-

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